

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Phase I Open-label, Single Dose Study to Investigate the Effect of Hepatic Impairment on M2951 (BTK Inhibitor) PK **Short Title/Acronym:** Effect of Hepatic Impairment on M2951 (BTK Inhibitor) PK **Protocol Number:** MS200527_0059 **NCT Number:** NCT04546789 **Sponsor Name:** Merck Healthcare KGaA, Darmstadt, Germany **Investigational Product:** M2951 (Evobrutinib / BTK Inhibitor) **Phase of Development:** Phase I **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the effect of mild and moderate hepatic impairment on the pharmacokinetics (PK) of M2951 (Evobrutinib) compared to participants with normal hepatic function. **Secondary Objectives:** To evaluate the safety and tolerability of a single oral dose of M2951 in participants with hepatic impairment and healthy controls, as well as protein binding parameters.

2.2 Background & Rationale M2951 (Evobrutinib) is an orally active, highly selective Bruton's Tyrosine Kinase (BTK) inhibitor developed for autoimmune diseases like Multiple Sclerosis. Because hepatic function is a critical component of drug metabolism and clearance, this Phase I study evaluates how hepatic impairment (mild to moderate) influences the PK profile of M2951 to inform potential dose adjustments in these specific patient populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel-group (overtly healthy participants with normal hepatic function serving as the control) **Blinding:** Open-label **Randomization:** N/A (Assigned based on hepatic function status)

3.2 Planned Enrollment & Duration Target \$N\$: 24 participants **Number of Sites:** Global Phase I Clinical Pharmacology Units **Estimated Study Start:** 30-Sep-2020 **Estimated Primary Completion:** 29-Jun-2021

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with normal hepatic function (overtly healthy) OR participants with mildly impaired hepatic function (Child-Pugh class A, score 5 to 6, with confirmed liver cirrhosis) OR moderately impaired hepatic function (Child-Pugh class B, with confirmed liver cirrhosis), who have been clinically stable for at least 1 month prior to Screening. 2. Body weight within 50.0 and 120.0 kg and body mass index (BMI) within the range of 19.0 to 36.0 kg/m². 3. Female participants must not be pregnant or breastfeeding, or must not be a woman of childbearing potential (WOCBP).

4.2 Exclusion Criteria 1. Clinical history of autoimmune disorder with hepatic influence (though Hashimoto thyroiditis and rheumatic diseases are permitted) or any history of malignancy. 2. Diseases and surgeries of the gastrointestinal tract that could influence anatomy and mobility (e.g., prior cholecystectomy or inflammatory bowel disease, or any clinically relevant surgery within 6 months prior to Screening). 3. Primary and secondary biliary cirrhosis, clinical evidence of severe ascites, or Hepatic encephalopathy Grade > 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single oral dose of 30 mg M2951 (administered as 3 film-coated tablets of 10 mg). **Route of Administration:** Oral, taken on Day 1 after a standard breakfast. **Duration of Treatment:** Single dose (1 Day).

5.2 Concomitant & Prohibited Therapy Permitted: Routine care and stable medications for underlying hepatic cirrhosis comorbidities not interacting with study drug PK. **Prohibited:** Strong CYP inhibitors/inducers, or any medication known to alter gastric pH or gastrointestinal motility within predefined wash-out periods prior to dosing.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to -1	Informed Consent, Medical History, Hepatic Function Labs (Child-Pugh Scoring)
Baseline/Admission	Day -1 to Day 0	Admission to clinical facility, baseline safety assessments
Intervention	Day 1	Single Dose Administration of M2951 (30 mg), Intensive PK blood sampling
End of Study	Follow-up	Final PK labs, Vital Signs, AE monitoring, Unit Discharge

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Key Pharmacokinetic (PK) parameters, including Area Under the Concentration-time curve (AUC_{0-inf}, AUC_{0-t}), extrapolated AUC (AUC_{extra}), apparent terminal rate constant (λ_z), terminal half-life (T_{1/2}), and Maximum Plasma Concentration (C_{max}). **Measurement Method:** Calculated from serial plasma concentrations using log-linear regression analysis of the terminal log-linear phase.

7.2 Secondary & Exploratory Endpoints 1. Evaluation of unbound drug fraction (f_u), defined as the ratio of unbound drug concentration to total drug concentration (e.g., AUC_{0-inf,u}). 2. Safety and tolerability endpoints (incidence and severity of Adverse Events [AEs]).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical pharmacology unit monitoring including vital signs, ECG, laboratory values, and direct patient questioning throughout the inpatient stay.

Serious Adverse Events (SAEs): Reported within 24 hours to the sponsor's pharmacovigilance team. **Data Monitoring Committee**

(DMC): Internal safety review overseen by the principal investigator and Merck Medical Monitor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Analysis Set (all participants who received the drug and provided sufficient concentration data) and Safety Analysis Set (all participants who received at least one dose). **Sample Size Justification:** A sample size of N=24 is typically adequate for estimating differences in PK parameters (C_{max} and AUC) across defined hepatic function groupings.

Handling of Missing Data: Concentrations below the Lower Limit of Quantification (LLQ) were handled based on predefined PK guidelines (e.g., C_{last} pred evaluation).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Direct on-site monitoring within the Phase I clinical pharmacology facility to verify 100% of source data for primary endpoints. **Audit Trail:** Validated eCRF entries with audit trails and automated data clarification systems.

11. Study Identifiers & Governance

Primary ID: MS200527_0059 **Registry Number:** NCT04546789 **Sponsor/Lead Organization:** Merck Healthcare KGaA, Darmstadt, Germany
Study Title: Phase I Open-label, Single Dose Study to Investigate the Effect of Hepatic Impairment on M2951 (BTK Inhibitor) PK
Official Regulatory Title: Phase I Open-label, Single Dose Study to Investigate the Effect of Hepatic Impairment on M2951 (BTK Inhibitor) PK

12. Clinical Framework (Hepatic Impairment Specific)

Primary Condition: Hepatic Impairment (Child-Pugh Class A & B)
Diagnosis Threshold: Child-Pugh Scoring system (score 5-6 for mild, Class B for moderate) **Medical Methodology:** Single oral dose pharmacokinetics and safety monitoring **Optimization Target:** Dosage determination and clearance assessment of BTK Inhibitor M2951 in renally/hepatically impaired populations

13. Study Procedures & Methodology

Management Pathway: Inpatient Phase I trial clinical pathway
Education Protocol (HCP): Site initiation visits and Phase I unit protocol adherence training **Education Protocol (Patient):** Informed consent process, instructions on standardized meals (standard breakfast prior to dosing) **Quality Audit Mechanism:** Independent auditing by the sponsor's Quality Assurance department

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6-8 weeks (from Screening to End of Follow-up) **Onsite Visit Cadence:** Extensive inpatient monitoring during the dosing period (Day 1) and scheduled outpatient follow-up visits. **Remote Monitoring Frequency:** N/A for intensive Phase I PK tracking. **Checklist Review Interval:** Daily during the inpatient confinement phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				